Compound Interest, Payments and Loans

Concept

Compound interest grows a changing balance. Payment timing and rate periods must match.

Compound growth

$A = P(1 + r/m)^{mt}$; periodic rate $i = r/m$.

Deposits / loans

$B_{new} = B_{old}(1 + i) +$ deposit, or minus payment.

Regular payments

Equal end-period payments, $i>0$: $FV = D[(1+i)^{N} - 1]/i$; loan $R = Pi/[1-(1+i)^{-N}]$.

Compare costs

Check fees, timing, total payments and retained value. Rates here are hypothetical.

Example 1

Grow $2000 at 6% nominal annual interest, compounded monthly, for 2 years.

Solution

1. $i = 0.06/12 = 0.005$; $N = 24$.
2. $A = 2000(1.005)^{24} \approx \$2254.32$.
3. Interest $\approx \$254.32$.

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Example 2

A $1000 loan charges 1% monthly. Make a $90 end-of-month payment.

Solution

1. Interest $= 1000(0.01) = \$10$.
2. New balance $= 1000 + 10 - 90 = \$920$.
3. Principal repaid $= 90 - 10 = \$80$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the value of $1500 after 3 years at 4% compounded annually.

2. Find the effective annual rate for 12% nominal compounded monthly.

3. A $500 balance grows 2% in a period before a $60 payment. Find the new balance.

4. Why might doubling the loan term fail to halve the total borrowing cost?

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Practice answers | check after trying

1. $1500(1.04)^3 = \$1687.30$ to cents.

2. $(1.01)^{12} - 1 \approx 0.126825 = 12.68\%$.

3. $500(1.02) - 60 = \$450$.

4. Interest continues to accumulate on outstanding balances for longer; lower periodic payments do not necessarily mean lower total interest.